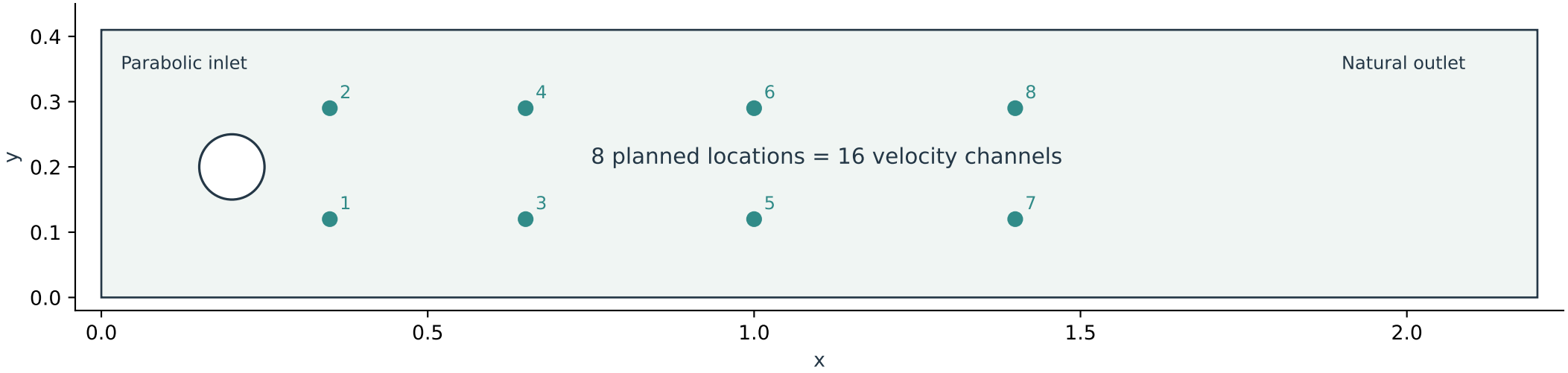
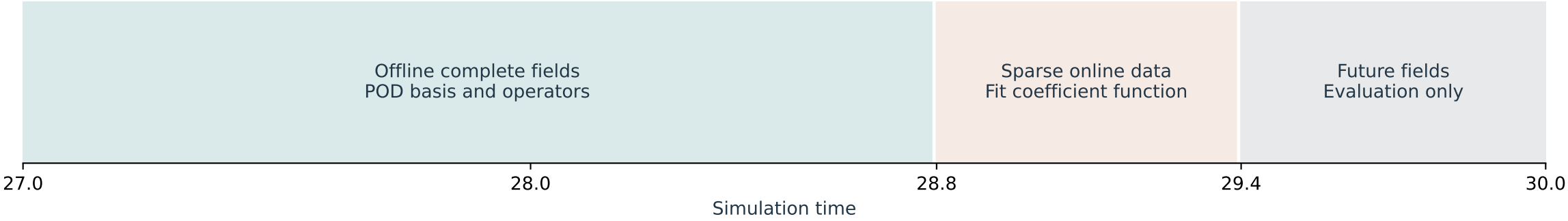
PINN extension · Phase 0 design schematic, not a trained-model result

A · Planned sparse observations in the cylinder channel (not optimized)



B · Separate the offline prior, online observations, and future evaluation



C · A coefficient-function PINN constrained by the reduced FEM equations

$$\widehat{u}(t) = \overline{u} + \Phi D z_\theta(\tau), \quad \tau = (t - t_A)/T_A$$

Fixed offline basis and mean
Same sensor information for all baselines

$$r_z = \partial_\tau z_\theta - T_A D^{-1} M_r^{-1} F_r(D z_\theta; \nu)$$

Initial state inferred from sparse data
No complete online truth in the loss
No future observations or collocation

$$\mathcal{L} = \mathcal{L}_{\text{sensor}} + \lambda_f \mathcal{L}_{\text{physics}}$$